The nightly reports already carry the warning signs; the hard part is separating them from broken data, weather and chronic noise

Harbor Table Foods, Plants A–C. June 2026 is real controller data; Apr–May and Jul–Sep are simulated, with operations records. Team SEC02_Contextual_Odyssey.

715,440
hourly readings, 88.9% usable, 3 plants, Apr–Sep 2026

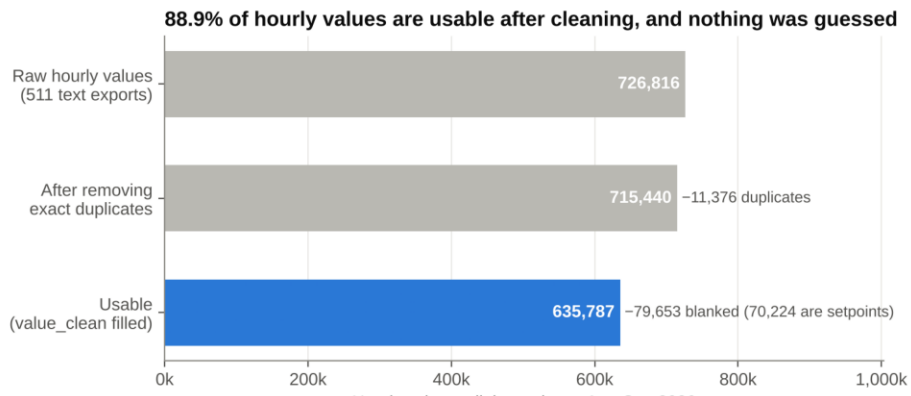
117
measured sensors (+38 setpoints)

309/309
review numbers traced to raw data (2 text errors)

26.6
alarms per week today (target: ≤2 per plant)

26
known events to score detection against

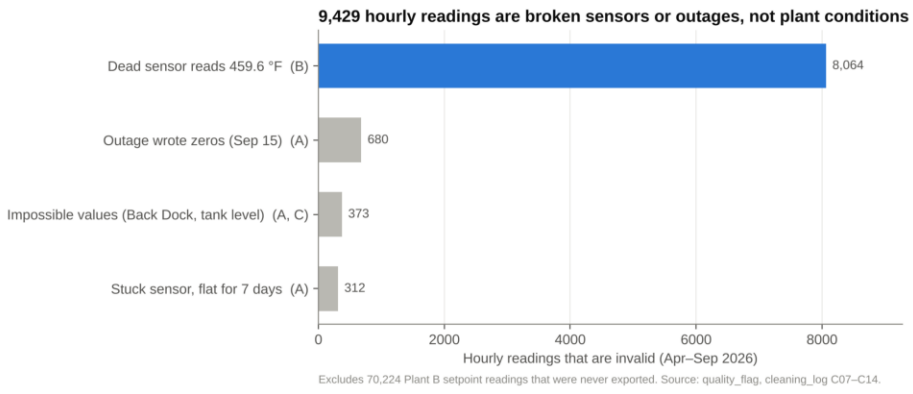
FROM RAW TO CLEAN DATA



	Raw	Clean
Controller files	511 .txt	12 CSVs
Point-blocks	30,284	29,810
Hourly values	726,816	715,440
Usable hourly values	**not separable**	**635,787 (88.9%)**
Exact duplicates removed		11,376
Blanked, kept in value_raw		79,653 (11.1%)
of which setpoints		70,224
Operations rows	26,440	26,440
Unparsed lines		0

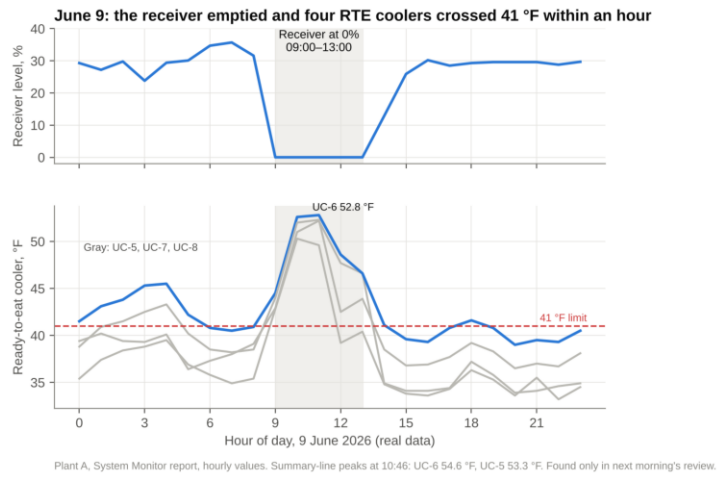
Takeaway	88.9% of hourly values are usable. Most of the 11.1% that is blank is Plant B setpoints that were never exported, not lost sensor readings. Only exact duplicates were deleted.
In plain terms	727k readings exported; 11k exact copies removed; 80k marked unusable (most are setpoints never exported).
Why it matters	89% is good data, nothing was guessed, and every original value is kept. Plant B's high blank rate is missing setpoints, so its normals must be learned from history.

DATA QUALITY



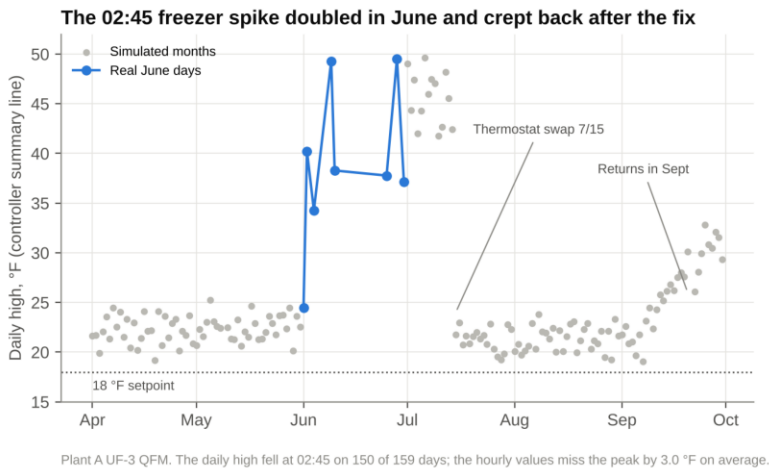
Takeaway	Plant B's dead sensor alone produces 8,064 readings of 459.6 °F. Flag these before any rule runs, or they fire as emergencies.
In plain terms	One disconnected sensor says 459.6 °F every hour; another was stuck for a week; an outage wrote zeros.
Why it matters	Believed as-is, these page people for nothing. Spot broken sensors first.

SUDDEN EVENTS



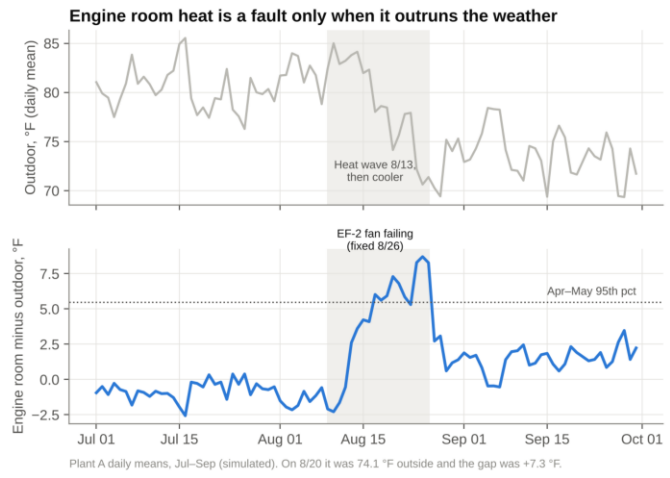
Takeaway	The receiver read 0% for 5 hours and discharge pressure fell 154→77 psig. A same-day rule would have fired at 09:00, not the next morning.
In plain terms	At 9 a.m. the ammonia tank ran empty; within an hour all four food coolers passed 41 °F.
Why it matters	A real problem seen only the next morning. An automatic check would flag it at 9 a.m.

SLOW TRENDS



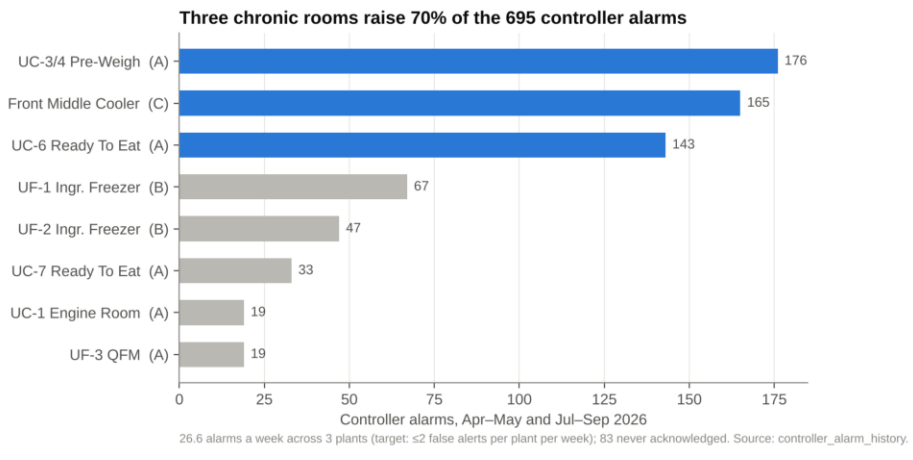
Takeaway	UF-3's daily high jumped from ~22 °F to 40–49 °F in June. Hourly values miss the 02:45 peak by 3.0 °F, so trend rules must read the summary line.
In plain terms	A nightly 2:45 a.m. warm-up doubled in June, was fixed in July, then crept back in September.
Why it matters	No single night looks bad. You only catch it by watching the trend over days.

CONTEXT: WEATHER



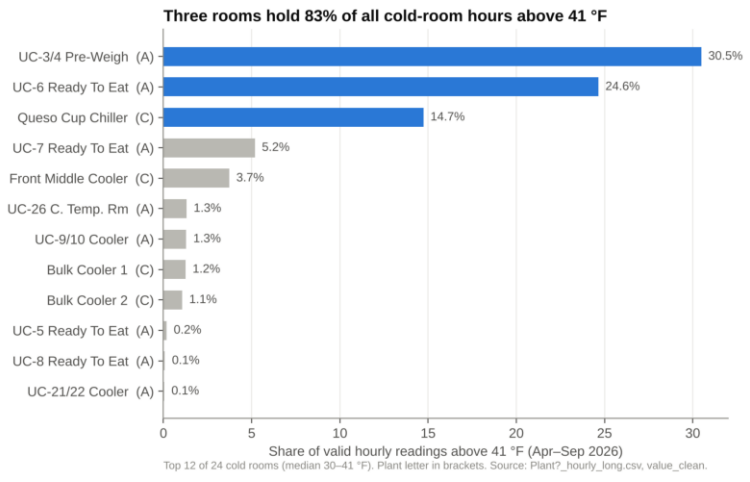
Takeaway	The hottest outdoor day (8/13) had a normal gap (+2.6 °F). The fan fault showed on cooler days (+7.3 °F on 8/20). Model the gap, not the raw temperature.
In plain terms	In the heat wave the gap stayed normal. On cooler days it jumped: a failing exhaust fan.
Why it matters	Watch the gap, not the raw temperature, or heat waves cause false alarms.

CONTEXT: FALSE ALARMS



Takeaway	26.6 alarms a week, 70% from three chronic rooms, 83 never acknowledged. Meanwhile 6 of 12 logged faults never alarmed on their own point.
In plain terms	About 27 alarms a week, mostly from three always-warm rooms; 83 were never acknowledged.
Why it matters	“Boy who cried wolf”: real problems, like a possible leak, never alarmed at all.

NORMAL BEHAVIOR



Takeaway	Across 24 cold rooms, 3.5% of hours are above 41 °F, and three rooms hold 83% of them. 'Normal' learned from history would hide these chronic rooms.
In plain terms	Most cold rooms almost never pass 41 °F; three rooms account for most of the time over it.
Why it matters	Risk is concentrated. Keep a fixed 41 °F rule so “always warm” is not treated as normal.

OVERALL TAKEAWAY

The sudden and slow faults we checked are visible in data the plants already export. A detector that (1) flags bad data first, (2) learns normal per point and hour, (3) reads the 15-minute summary lines for slow trends and (4) explains readings with weather, schedules and work orders can beat a 27-alarm-a-week status quo.

Sources: data/cleaned_v2 (controller hourly and daily summaries, operations records, cleaning documentation). Code: 01_exploratory_analysis/eda_deliverable/.